

33 2025-05-01 | Week 15 | Lecture 33

33.1 Simple linear regression (i.e., finding a best-fit line)

Based on chapter 12.1 in the textbook

The nexus question of this lecture: How do we find the “best fit” line for data?

Suppose we have four data points taking the form $(t, b) = (\text{time}, \text{observation})$:

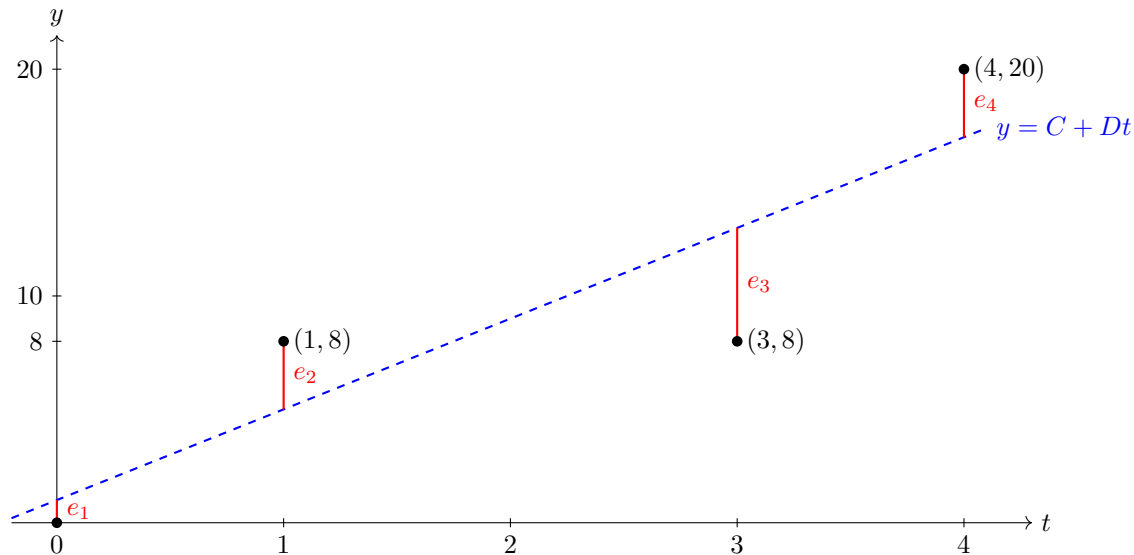
$$\begin{aligned}(t_1, b_1) &= (0, 0) \\ (t_2, b_2) &= (1, 8) \\ (t_3, b_3) &= (3, 8) \\ (t_4, b_4) &= (4, 20)\end{aligned}$$

I have plotted these below. The equation of a line is

$$y = C + Dt.$$

Definition 145. The *simple linear regression problem* is to find values for C and D which best fit our data.

Here’s a picture of the data points and a general line $y = C + Dt$ which aims to fit to the data:



Let $e_i = b_i - (C + Dt_i)$ for each $i = 1, \dots, 4$. In words, e_i is obtained by measuring the vertical distance between the line $y = C + Dt$ and the i^{th} data point. For this example, we have

$$\mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{bmatrix} = \begin{bmatrix} 0 - C \\ 8 - (C + D) \\ 8 - (C + 3D) \\ 20 - (C + 4D) \end{bmatrix}$$

This vector is called the *residual vector*, and it consists of the “errors” between

1. the observed data; and,
2. the values predicted by the line $y = C + Dt$.

We want all four errors $e_1, \dots, e_4$ to be small. One way to measure the total error is to square the four errors and add them up:

$$\text{SSE} = e_1^2 + e_2^2 + e_3^2 + e_4^2 \quad (25)$$

This is called the **sum of squared errors** (SSE). Squaring has two advantages: it makes all the errors positive, and it penalizes large errors.

Therefore, to find the best-fitting line $y = C + Dt$, our goal will be to find values for C and D which minimize the sum of squared errors $e_1^2 + e_2^2 + e_3^2 + e_4^2$.

We'll present two approaches to finding the best values for C and D .

Approach #1: Calculus

Consider the function

$$\begin{aligned} f(C, D) &= e_1^2 + e_2^2 + e_3^2 + e_4^2 \\ &= C^2 + (8 - C - D)^2 + (8 - C - 3D)^2 + (20 - C - 4D)^2. \end{aligned}$$

We can use calculus to minimize f . To find the critical point of f , we set $\nabla f = 0$ and solve for C and D :

$$0 \stackrel{\text{set}}{=} \frac{\partial f}{\partial C} = 2C - 2(8 - C - D) - 2(8 - C - 3D) - 2(20 - C - 4D) = -72 + 8C + 16D$$

and

$$0 \stackrel{\text{set}}{=} \frac{\partial f}{\partial D} = -2(8 - C - D) - 6(8 - C - 3D) - 8(20 - C - 4D) = -224 + 16C + 52D.$$

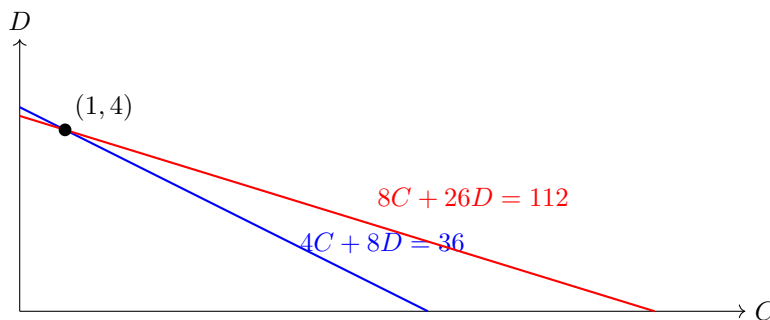
Combining these equations gives

$$\begin{cases} 8C + 16D = 72 \\ 16C + 52D = 224 \end{cases}.$$

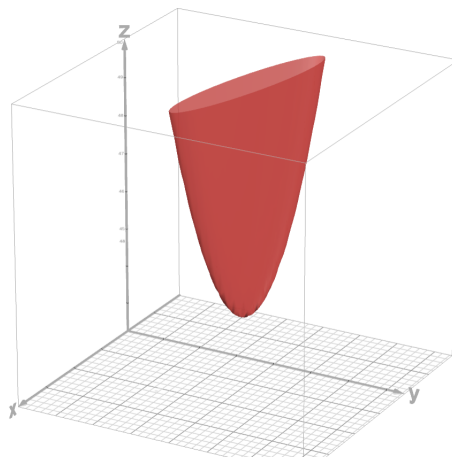
After dividing both equations by 2, we obtain the equivalent system

$$\begin{cases} 4C + 8D = 36 \\ 8C + 26D = 112 \end{cases} \quad (26)$$

This system corresponds to two lines which intersect at the point $(C, D) = (1, 4)$ (shown below) which is the critical point.



We need to check that $(C, D) = (1, 4)$ is a minimum. We could show this with the second derivative test, but there is also a nice geometric reason why it must be a minimum. In particular, observe that f is a sum of squares, so it is bounded below by zero. By considering cross-sections (i.e., freeze one of either C or D and vary the other), we get an upward-opening parabola in any direction. So the graph is an upward-opening elliptic paraboloid; i.e., a bowl-shape, as shown:



Hence f has exactly one critical point and the critical point is the global minimum.

Therefore f is minimized at $(C, D) = (1, 4)$. Hence, the best fitting line for our data is

$$y = 1 + 4t,$$

as this line, out of all possible lines, minimizes the sum of squared errors $e_1^2 + e_2^2 + e_3^2 + e_4^2$.

Approach #2: Linear Algebra

There is a compact way to organize these computations using matrices, which allows us to compute the best fit line without taking any derivatives.

To motivate this, observe that ideally our best-fit line would pass through all the points, which occurs precisely when C and D satisfy the following system of equations:

$$\begin{cases} C + 0 \cdot D = 0 \\ C + 1 \cdot D = 8 \\ C + 3 \cdot D = 8 \\ C + 4 \cdot D = 20 \end{cases}$$

This is a system of 4 linear equations in 2 unknowns (C, D) . We can write it in matrix form as

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}}_A \underbrace{\begin{bmatrix} C \\ D \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}}_b$$

or more compactly as

$$Ax = b.$$

The bad news is that there is no choice of C and D for which this equation is true (because the data points **do not lie on a line**).

So instead we want Ax be as close as possible to b ; that is

$$Ax \approx b.$$

Since $\mathbf{e} = b - Ax$, we want

$$\mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{bmatrix} \approx 0.$$

So we want the vector $\mathbf{e}$ to be as tiny as possible. By minimizing $e_1^2 + e_2^2 + e_3^2 + e_4^2$, we will also simultaneously minimize $\sqrt{e_1^2 + e_2^2 + e_3^2 + e_4^2}$, which is the length of $\mathbf{e}$. This gives a geometric interpretation of our goal:

finding the values of C and D which give the best fit line $y = C + Dt$ to our data is equivalent to finding the values of C and D which **minimize the the length of the residual vector $\mathbf{e}$** .

To do this, we multiply both sides of $Ax = b$ by

$$A^\top = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \end{bmatrix}.$$

This yields the following system of equations, called the *normal equations*:

$$A^\top Ax = A^\top b. \tag{27}$$

Writing out Eq. (27), we get

$$\begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

or equivalently,

$$\begin{cases} 4C + 8D = 36 \\ 8C + 26D = 112 \end{cases}$$

This system is the normal equations. Observe that it is the same as Eq. (26) and has the same solution: $(C, D) = (1, 4)$, which is the minimizer of $\|\mathbf{e}\|^2 = e_1^2 + \dots + e_4^2$.